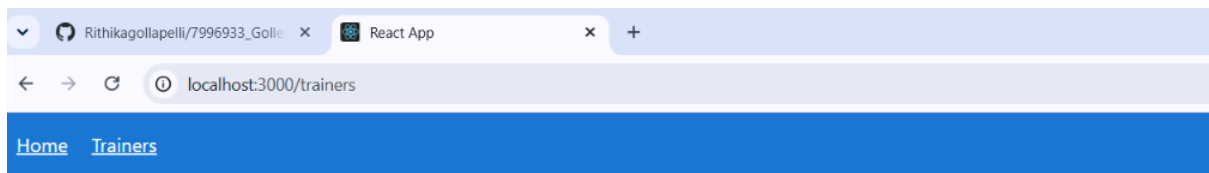
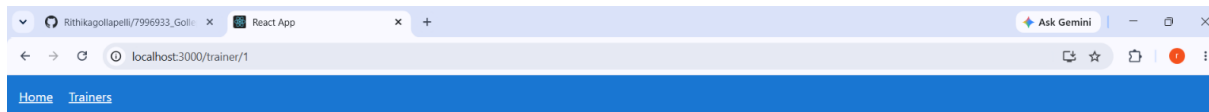


# Output6 trainersapp. ReactJS-HOL



## Trainer List

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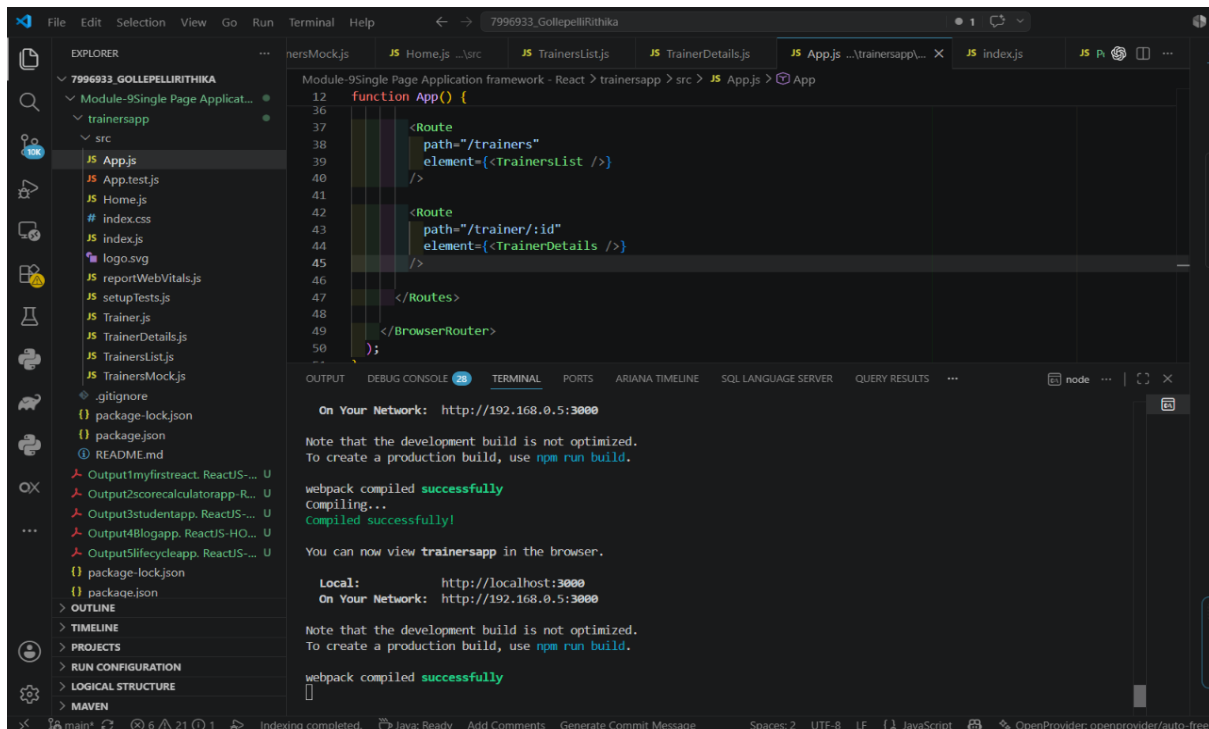
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## Objectives

### 1. Explain the Need and Benefits of React Router

React Router enables navigation between different pages in a React Single Page Application (SPA) without reloading the entire webpage. It improves the user experience by providing fast, seamless page transitions while maintaining application state. React Router also helps organize the application into multiple reusable components, making the code easier to develop, maintain, and scale.

### 2. Identify the Components in React Router

React Router provides several components that make navigation simple and efficient. Common components include:

- **BrowserRouter** – Wraps the entire application and enables routing using the browser's history.
- **Routes** – Holds all route definitions.
- **Route** – Maps a URL path to a specific React component.
- **Link** – Creates navigation links without refreshing the page.
- **NavLink** – Similar to Link but highlights the active navigation link.
- **Outlet** – Renders nested child routes within a parent route.
- **Navigate** – Redirects users from one route to another programmatically.

### 3. List the Types of Router Components

React Router supports different router components based on application requirements:

- **BrowserRouter** – Uses the browser's History API and is the most commonly used router for web applications.
- **HashRouter** – Uses the URL hash (#) for routing and is suitable for static hosting environments.
- **MemoryRouter** – Stores navigation history in memory and is mainly used for testing or non-browser environments.
- **StaticRouter** – Used for server-side rendering (SSR) applications.

#### 4. Parameter Passing via URL

React Router allows data to be passed through URL parameters, enabling dynamic routing. URL parameters are useful for displaying specific information such as user profiles, product details, or order information based on an identifier. These parameters are defined in the route using a colon (:) and accessed inside the component using the `useParams()` hook, making applications more dynamic and interactive.